



- **WEEK 3**
- **LINEAR PROGRAMMING - GRAPHICAL SOLUTIONS**
- **Represent constraints as inequalities**
- **Identify a feasible region (solution space)**
- **Find corner points**
- **Optimize an objective function (maximize or minimize)**

👉 **Key Idea:**

The best solution ALWAYS occurs at a corner point.

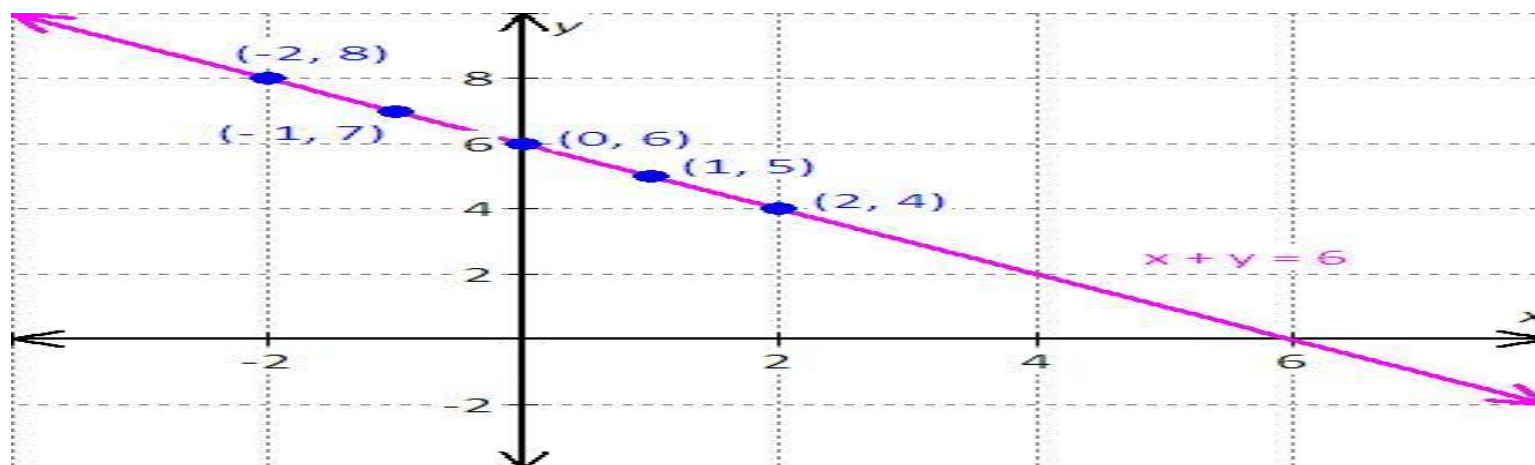
◆ **A. GRAPHING LINEAR INEQUALITIES**

Concept

To graph an inequality:

- 1. Replace inequality with equality (draw line)**
- 2. Use intercepts to plot the line**
- 3. Test a point (usually (0,0))**
- 4. Shade the correct region**

📊 **Example: Graph $x+y \leq 6$**



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Steps:

- Draw line: $x+y=6$
- Intercepts: $(0,6)$, $(6,0)$
- Test $(0,0)$: TRUE → Shade that side

Line Types

Inequality Line Type	Meaning
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Inequality	Line Type	Meaning
$\leq$ or $\geq$	Solid	Line included
$<$ or $>$	Dashed	Line excluded

Shading Rule

- $\leq$ or $<$ → Shade below/left
 - $\geq$ or $>$ → Shade above/right
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B. FEASIBLE REGION

Definition

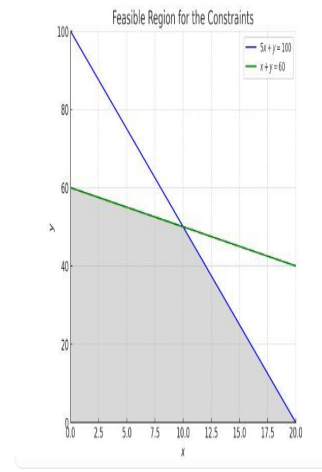
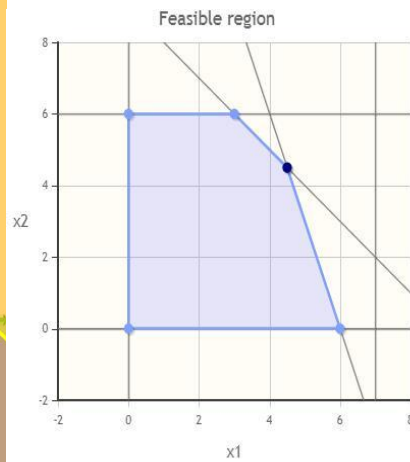
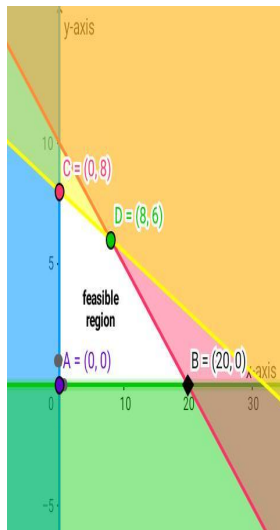
The feasible region is the area satisfying ALL constraints simultaneously.

Example:

Constraints:

- $x+y \leq 6$

- $2x + y \leq 10$
- $x \geq 0, y \geq 0$



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Corner Points:

- $A = (0, 0)$
- $B = (0, 6)$
- $C = (4, 2)$
- $D = (5, 0)$



👉 **Feasible region = polygon ABCD**

📊 **Finding Intersection (Example)**

Solve:

- $x+y=6$
- $2x+y=10$

Result:

- $x=4, y=2$



📌 **C. OPTIMIZATION (GRAPHICAL METHOD)**

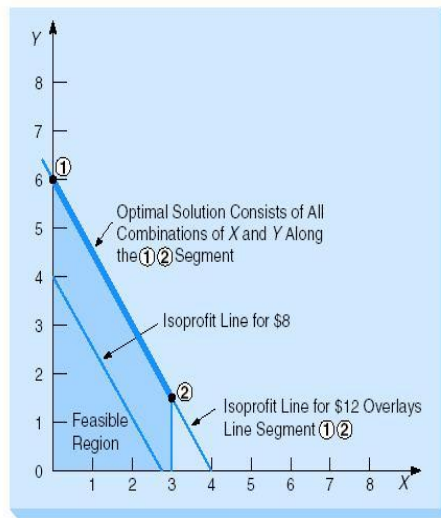
📌 **Corner Point Theorem**

👉 **Optimal solution occurs at a vertex (corner point)**

📌 **Steps:**

1. Graph constraints
2. Identify feasible region
3. Find corner points
4. Substitute into objective function
5. Choose max/min value

TYPES OF SOLUTIONS

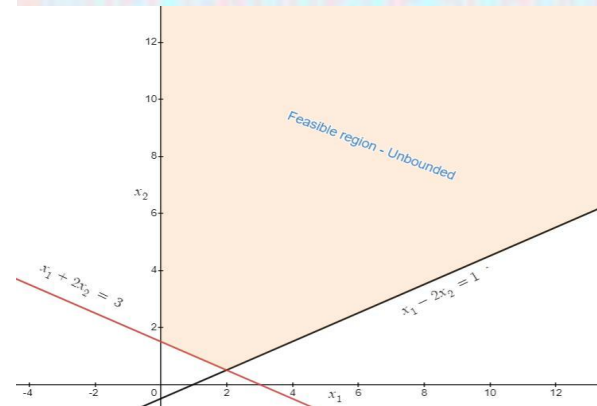


Multiple optimal solutions

□ For multiple optimal solutions to exist, the objective function must be parallel to one of the constraints

• Any weighted average of optimal solutions is also an optimal solution.

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1. Unique solution
2. Multiple solutions
3. Unbounded solution
4. No feasible solution

◆ **WORKED EXAMPLE 1**

Maximize:

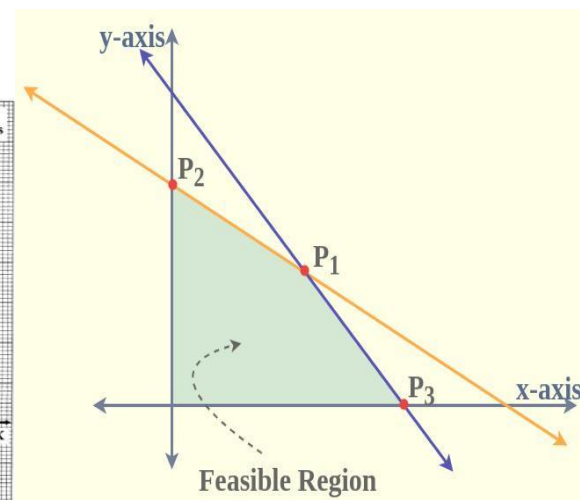
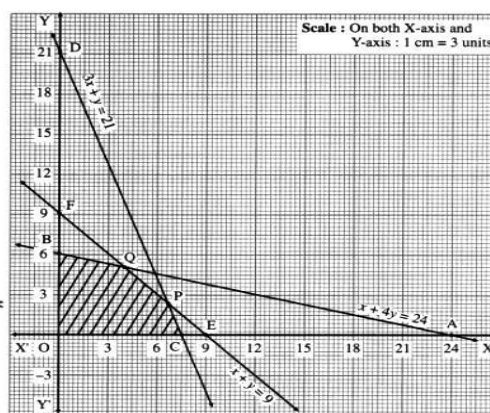
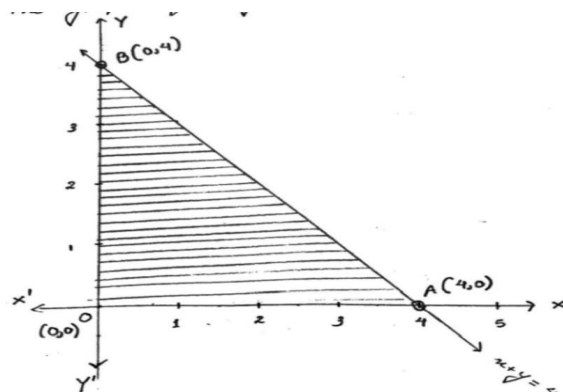
$$P=3x+4y$$

Subject to:

- $x+y \leq 6$
- $2x+y \leq 10$
- $x, y \geq 0$



▮ **Graph Solution**



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Evaluate:

Point Value

(0,0) 0

(0,6) 24 ✓

(4,2) 20

(5,0) 15

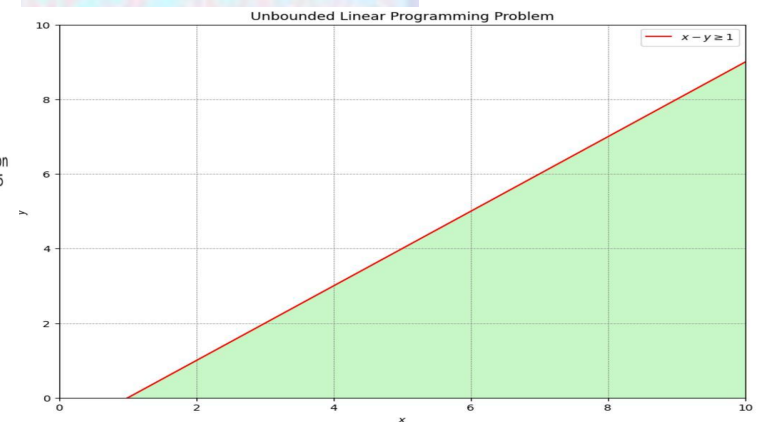
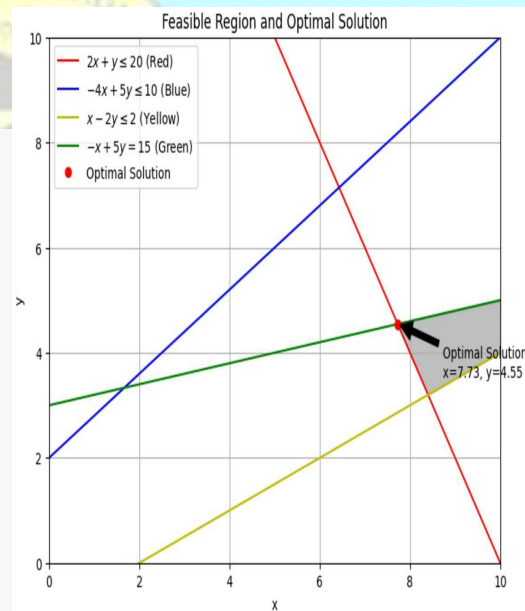
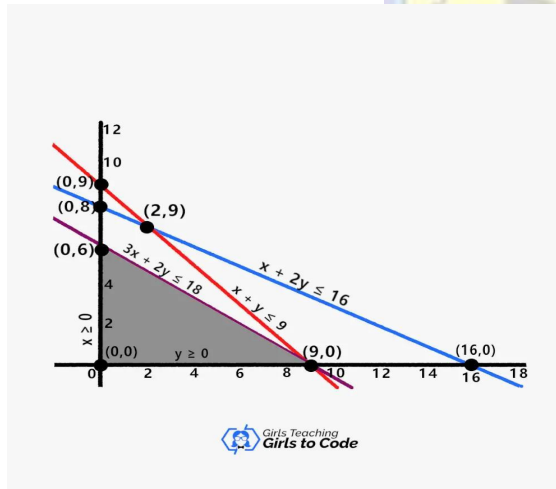
✓ **Answer:**

Maximum $P=24$ at $(0,6)$

WORKED EXAMPLE 2 (MINIMIZATION)

Minimize:

$$C=2x+3y$$





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Result:

Minimum = 10 at (2,2)

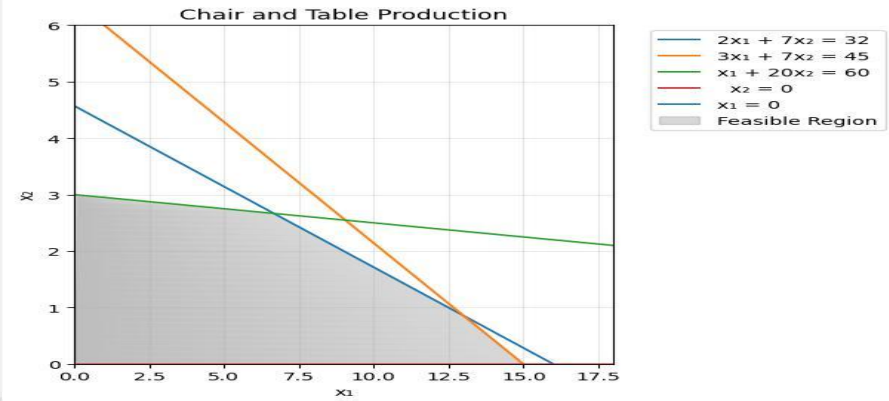
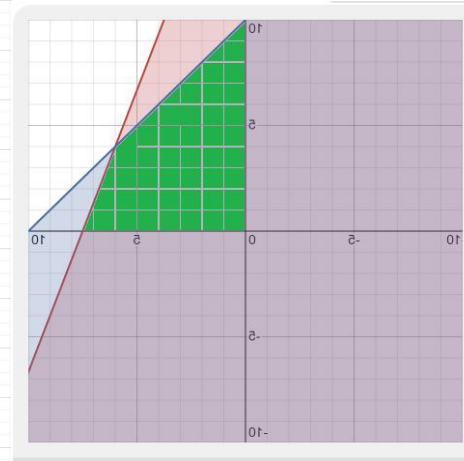
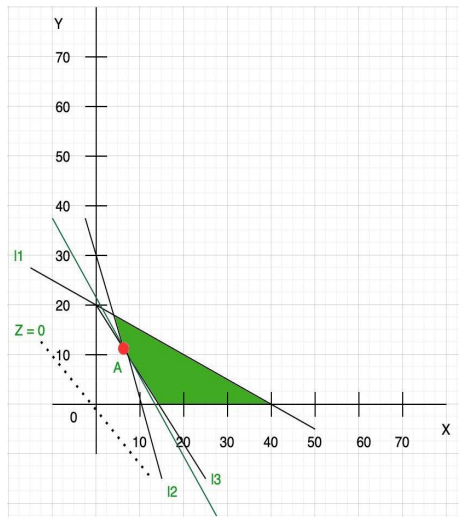
◆ **REAL-LIFE APPLICATION (FACTORY)**

Problem:

- **Chairs (x), Tables (y)**
- **Time + wood constraints**
- **Maximize profit**

▮ **Graph**





Answer:

👉 **Max profit = ₦80,000**

👉 **Produce:**

- **40 chairs**
- **0 tables**

🔑 CLASS ACTIVITIES



1. Graph: $x+2y \leq 8$

2. Find corner points:

- $x+y \leq 5$
- $2x+y \leq 8$

3. Evaluate objective function

? EVALUATION QUESTIONS

1. How do you choose shading side?

2. Define feasible region

3. State corner point theorem

4. Graph $2x+3y \leq 12$

5. Find vertices for given constraints

🏠 ASSIGNMENT (EXAM STANDARD)

1. Maximize:

$$P=5x+4y$$



2. Minimize:

$$C=3x+2y$$

3. Farmer Problem (LP modeling)

© FINAL SUMMARY (VERY IMPORTANT)

- **LP uses graphs to solve real-life problems**
- **Feasible region = overlap of all constraints**
- **Vertices determine optimal solutions**
- **Always test corner points only**

